

Application Layer & Security

Application Layer: Interfaces directly with applications and provides common network services.

DNS

Domain Name System:
Translates human-readable
domain names to IP addresses.

HTTP

Hypertext Transfer Protocol:
Used for fetching web
resources. Stateless protocol.

Cryptography

Securing communication in the
presence of adversaries.

Email Protocols:

- SMTP: For sending emails.
- POP3/IMAP: For retrieving emails.

Asymmetric Key Cryptography:

Different keys (Public and Private) used for
encryption and decryption (e.g., RSA). Solves key
distribution problem.

Symmetric Key Cryptography:

Same key used for encryption and decryption (e.g.,
DES, AES). Fast but requires secure key
distribution.

Firewalls:

Network security system that monitors and
controls incoming and outgoing network traffic
based on predetermined security rules.